

G8-03-CX-critical-functional-dynamic-kill-test-v1: Criticality + dynamics + bridge to W_+

ZFL-CX

September 3, 2026

1 Frozen input

$G7 = \text{FROZEN}, G8\text{-DESIGN-AUDIT} = \text{DONE}$	$G8\text{-01} = \text{KILLED}, G8\text{-02} = \text{KILLED}$	$\mathcal{C}_{\text{PROVEN}} = \{C_E^{\text{weak}}\}$
$\mathcal{C}_{\text{PROVEN}} \Rightarrow \perp \text{ OPEN} \quad \left[W_+ = \left(\int \omega^T S \omega \right)_+, \int_0^{T^*} W_+ = +\infty \text{ if } T^* < \infty \right]$		

2 Mission G8-03

Find critical X such that $\boxed{T^* < \infty \Rightarrow C_X}$ and $\boxed{C_X \wedge \mathcal{C}_{\text{PROVEN}} \Rightarrow \perp}$ or $\boxed{C_X \Rightarrow \int_0^{T^*} W_+ < \infty}$.

Concrete mechanism required: $\boxed{\frac{d}{dt}X + D_X \leq A_X, W_+ \leq cD_X + R_X, \int_0^{T^*} R_X < \infty}$.

3 Candidate X

Critical scalings: $\|u\|_{L^3}, \|S\|_{L^3}, \|\omega\|_{L^{3/2}}, \|u\|_{BMO^{-1}}, \|\omega\|_{\dot{B}_{3/2,\infty}^0}$. All invariant under $u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$. Pick $X(t) = \|S(t)\|_{L^3}^3 \sim \|\omega\|_{L^{3/2}}^{3/2} \sim \|u\|_{L^3}^3$ (same critical class). Evolution: from $D_t S = -S^2 - \Omega^2 - \nabla^2 p + \nu \Delta S$, $\frac{d}{dt}\|S\|_{L^3}^3 \sim \int |S| S : D_t S \sim - \int |S| S : (S^2 + \Omega^2 + \nabla^2 p) + \nu \int |S| S : \Delta S$. Dissipation $D_X \sim \nu \|\nabla |S|^{3/2}\|_2^2 \sim \nu \|\nabla S |S|^{1/2}\|_2^2$.

4 Attempted bridge

We need $W_+ = \int |\omega|^2 \alpha \leq cD_X + R_X$. $W_+ \lesssim \int |S| |\omega|^2 \lesssim \|S\|_{L^3} \|\omega\|_{L^3}^2$. By Sobolev, $\|\omega\|_{L^3} \lesssim \|\nabla \omega\|_{L^{3/2}} \lesssim \|S\|_{L^3}^{1/2} \|\nabla S\|_{L^2}^{1/2}$? Interpolation gives $\|S\|_{L^3}^3 \leq C \|S\|_{L^2}^{3/2} \|\nabla S\|_{L^2}^{3/2}$, not integrable without $\|S\|_{L^2}$ decay which is C_E^{weak} only $L_t^\infty L^2$ not L_t^2 .

Try $W_+ \leq \varepsilon D_X + C_\varepsilon \|S\|_{L^3}^5$ via Young: $ab \leq \varepsilon a^p + C_\varepsilon b^{p'}$. Then $R_X \sim \|S\|_{L^3}^5$, super-critical power $5 > 3$, $\int R_X$ not controlled by $\mathcal{C}_{\text{PROVEN}}$. Since C_{L^3} allows $\|S\|_{L^3} \sim \|u\|_{L^3} \rightarrow \infty$, R_X can diverge as $T^* \rightarrow \infty$.

BMO^{-1} attempt: $\|u\|_{BMO^{-1}}$ critical, but $W_+ \lesssim \|u\|_{BMO^{-1}} \|\nabla \omega\|_{L^2}^2$ again, with $\|\nabla \omega\|_2^2$ not integrable under $\mathcal{C}_{\text{PROVEN}}$ (only $\int_0^{T^*} \|\nabla u\|_2^2 < \infty$ from energy, but $\nabla \omega$ is H^1 not L^2).

Hence no identity $\frac{d}{dt}X + D_X$ dominates W_+ with integrable remainder. Criticality alone gives X diverges, but dissipation D_X is $\nu\|\nabla S\|_2^2$ scaling, while W_+ is cubic $S\omega^2 \sim S^3$, supercritical relative to D_X .

5 Kill criteria checked

1. X critical without useful dynamics? Yes — dynamics gives super-critical remainder. 2. $X \lesssim \|u\|_3$ no new info? Yes — $\|S\|_{L^3} \sim \|\nabla u\|_{L^3} \lesssim \|u\|_{L^3}^a \|\nabla^2 u\|^{1-a}$, returns to C_{L^3} . 3. Sobolev/CZ back to $\|u\|_3 \|\nabla \omega\|_2^2$? Yes. 4. Smallness assumed not derived from $T^* < \infty$? To close would need $\|S\|_{L^3} \ll 1$, not implied by blow-up.

6 Outcome

KILLED No triple connection criticality + temporal dynamics + quantitative bridge $W_+ \leq cD_X + R_X$, $\int R_X < \infty$. Any critical X gives $\frac{d}{dt}X + D_X \leq C\|S\|_{L^3}^5$ type, R_X not integrable under $\mathcal{C}_{\text{PROVEN}}$. $\mathcal{C}_{\text{PROVEN}} \not\perp$ remains OPEN. G8-01, G8-02, G8-03 all KILLED without dimensional estimate as bridge.